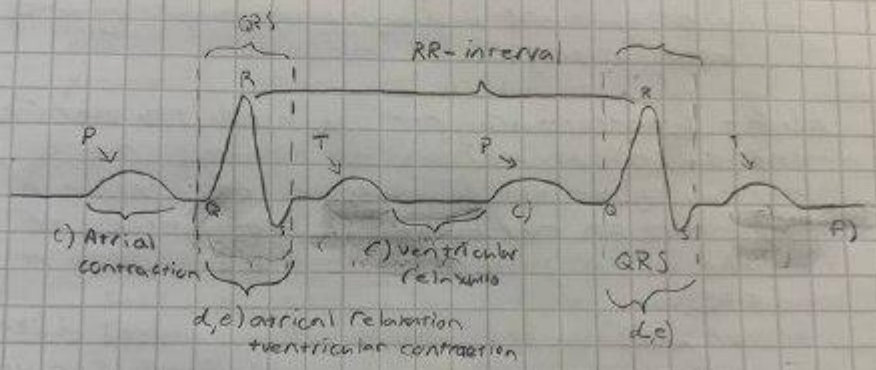
Weekly exercise 1:

Processing of Biosignals

Weekly exercise I:

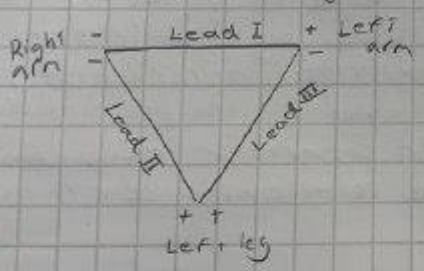
1) Biosignals are signals that can be measured from biological beings. So Biosignals reflect the nature and activities of physiological processes and systems in the body. With biosignals we can find abnormalities and diagnose diseases.

2)



3)

Einthoven's triangle:



Einthoven's triangle is an imaginary formation of three limb leads. Triangle includes three electrodes, which are located to right arm, left arm and left leg. Those electrodes are creating three leads.

Connecting these leads mathematically, we can create more leads, so we are getting wider information about heart.

The electrical impulses of the heart can be measured by voltages between these three electrodes. The triangle allows to measure the direction too.

4)

For example to epilepsy diagnosing and sleep disorder analysis. We need computer tools because EEG-data includes a huge amount of data and the data is much more complex than ECG-data.
With computers we can eg filter out the data we don't need.

5) EMG - Electromyogram is the bioelectrical measurement of muscle contractions. The activity of measured muscles is visible as increased voltage between the electrodes.
- EMG can aid in diagnosing muscle injuries and strength of muscles.

ENG - Electroneurogram is used to measure the electrical activity of neurons in the nervous system. It is observed by applying a stimulus an measuring the propagation of the action potential over the length of a nerve
- With ENG we can find various neural diseases by researchin latencies an a nerve conduction velocity.

PPG - Photoplethysmogram is used to detect blood volume changes in the tissues by illuminating the skin with LED and measuring the reflected or transmitted light as an output on the photodiode.
- PPG can be used to oxygen saturation measurement and to detect premature ventricular contraction (PVC)

6)

- Frequency analysis: Biosignals include many different frequencies. These can be revealed for example with Fourier Transform.
  - For example EEG include alfa, beata, theata and delta bans

- Amplitude: Amplitude describes the strength of the signal

- Temporal dynamics: Time domain analysis involves examining biosignals in the time dimension, providing information about the behavior over time
  - In EMG the duration of musele contraction and in EKG durations of intervals

- Statistical measures: Statistical properties such as mean and variance offer quantitive information into the distribution and characterize the complexity of biosignals.